



Review Test Submission: Assignment Week 5

User Mr W (Wouter) W Nel

Course EBN 122 S2 2017

Test Assignment Week 5

Started 8/26/17 2:16 PM

Submitted 8/28/17 6:14 PM

Due Date 8/28/17 11:59 PM

Status Completed

Attempt Score 13.5 out of 20 points

Time Elapsed 51 hours, 58 minutes

Instructions Read over the document *ClickUP Assignments Instructions*.

Before sending an enquiring regarding ClickUP question to any-one please verify that your problem is valid ... i.e. if you suspect incorrect marking first check with a fellow student or two.

Please direct any enquiries regarding ClickUP assignments to EBN2017.ClickUptest@gmail.com

Save the answer to each question as you complete it.

At the end of the assignment click Save and Submit.

Give your final correct answer 3 significant digits.

Make sure you use the FULL STOP for your decimal place marker.

You must indicate the units, unless otherwise specified:

- There must be NO space between the answer and the units
- There must be between 1 and 3 digits to the left of the decimal point. *This means that you need enter 1kV rather than 1000V and 100mV rather than 0.1V*
- Write out 'ohm' to indicate Ω (use singular)
- Use 'u' to indicate micro.
- Only use pico(**p**), nano(**n**), micro(**u**), milli(**m**), kilo(**k**), mega(**M**), giga(**G**) and tera(**T**); do not use centi, deci and deca

Results All Answers, Submitted Answers, Correct Answers, Feedback, Incorrectly Answered Questions

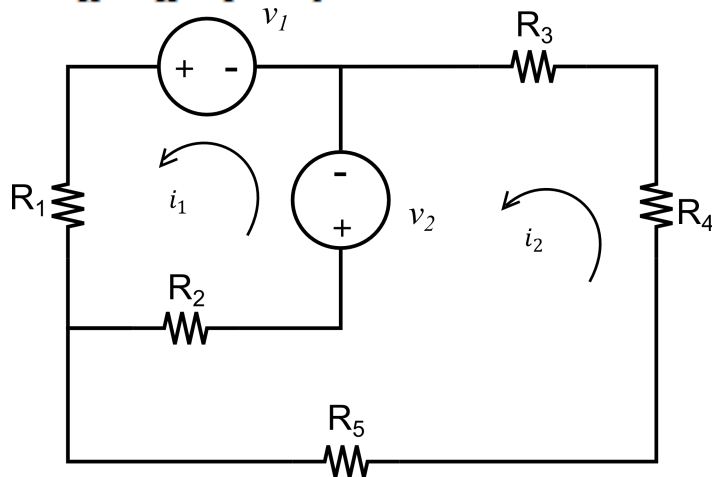
Question 1

2 out of 2 points



Use inspection to set up a matrix equation in terms of the mesh currents.
 $R_1=93\Omega$, $R_2=10\Omega$, $R_3=36\Omega$, $R_4=17\Omega$, $R_5=88\Omega$, $v_1=2V$ and $v_2=2V$

In $\begin{bmatrix} R_{11} & R_{12} \\ R_{21} & R_{22} \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \end{bmatrix} = \begin{bmatrix} V_a \\ V_b \end{bmatrix}$, give the value for V_b



Selected Answer: ☒ 2

Correct Answer: ☒ 2 ± 0

Question 2

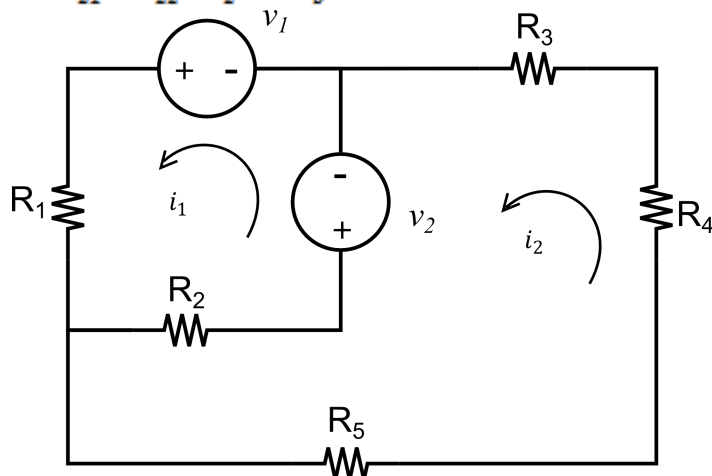
2 out of 2 points



Use inspection to set up a matrix equation in terms of the mesh currents.

$R_1=59\Omega$, $R_2=89\Omega$, $R_3=55\Omega$, $R_4=87\Omega$, $R_5=99\Omega$, $v_1=2V$ and $v_2=5V$

In $\begin{bmatrix} R_{11} & R_{12} \\ R_{21} & R_{22} \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \end{bmatrix} = \begin{bmatrix} V_a \\ V_b \end{bmatrix}$, give the value for R_{22}



Selected Answer: ☒ 330

Correct Answer: ☒ 330 ± 0

Question 3

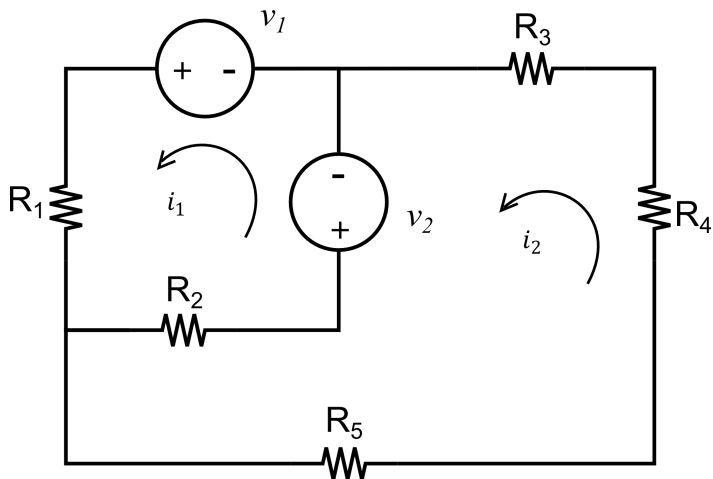
2 out of 2 points



Use inspection to set up a matrix equation in terms of the mesh currents.

$R_1=30\Omega$, $R_2=16\Omega$, $R_3=81\Omega$, $R_4=82\Omega$, $R_5=54\Omega$, $v_1=8V$ and $v_2=6V$

In $\begin{bmatrix} R_{11} & R_{12} \\ R_{21} & R_{22} \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \end{bmatrix} = \begin{bmatrix} V_a \\ V_b \end{bmatrix}$, give the value for R_{12}



Selected Answer: ☒ -16

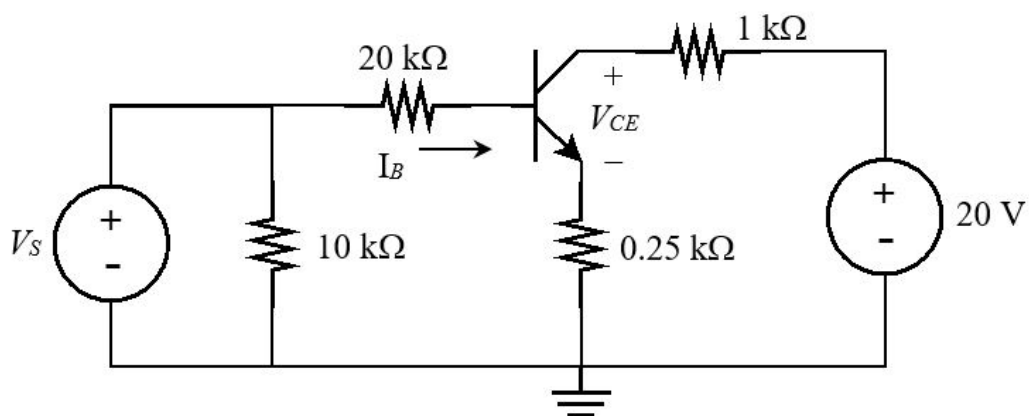
Correct Answer: ☒ -16 ± 0

Question 4

2 out of 2 points



With $I_B = 409 \mu\text{A}$ and $\beta = 99$ determine V_{CE} .



Selected Answer: ☒ -30.7V

Correct Answer: ☒ $-30.72 \pm 5\%$ (V)

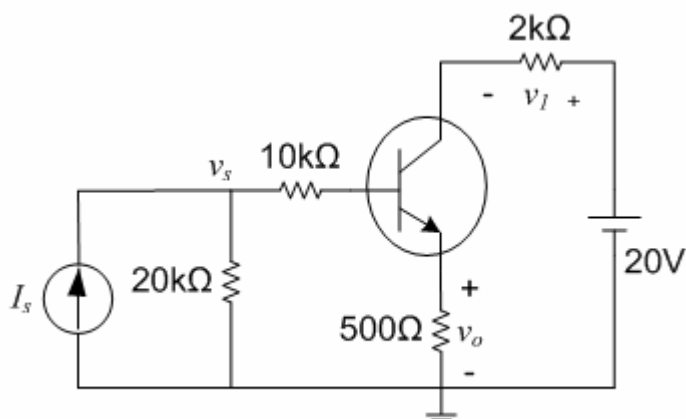
Question 5

0.5 out of 2 points



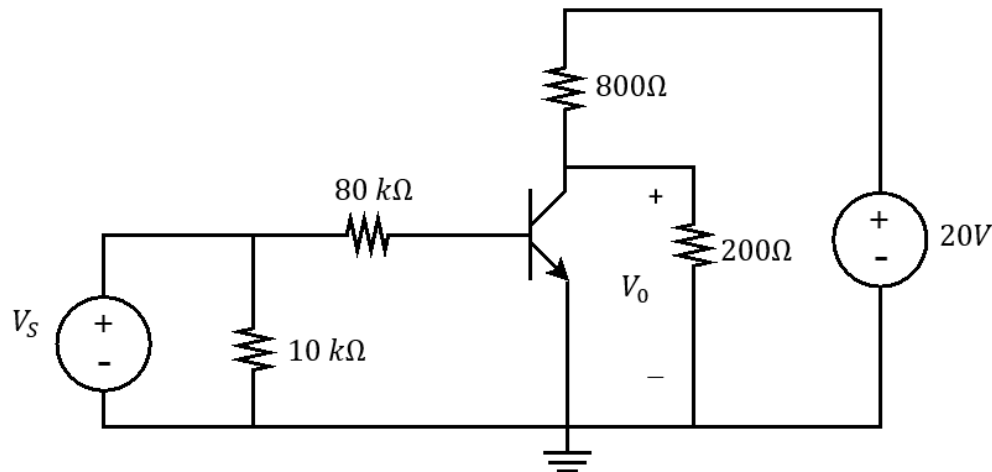
$\beta = 100$ and $V_{BE} = 0.7 \text{ V}$

Determine V_S if $V_O = 7 \text{ V}$.

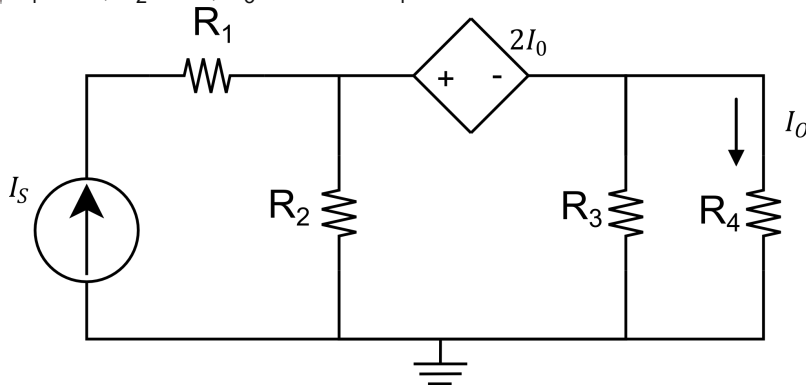


Selected Answer:  -9.09VCorrect Answer:  $9.09 \pm 5\%$ (V)**Question 6**

0.5 out of 2 points

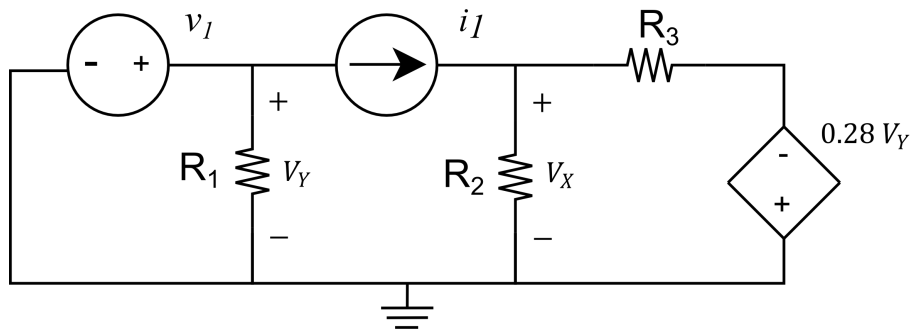
Given: $\beta = 100$, $V_{BE} = 0.7V$ and $V_S = 6V$, find V_O .Selected Answer:  -26.7VCorrect Answer:  $2.53 \pm 5\%$ (V)**Question 7**

0 out of 2 points

Using the principle of linearity, determine I_O/I_S  $R_1=74\Omega$, $R_2=65\Omega$, $R_3=12\Omega$ and $R_4=98\Omega$ Selected Answer:  93.4mCorrect Answer:  $0.093 \pm 2\%$ **Question 8**

0.5 out of 2 points

Determine the individual contribution of the v_i current source to V_X . $R_1=12\Omega$, $R_2=69\Omega$, $R_3=44\Omega$ and $v_i=16V$ and $i_i=108mA$.



Selected Answer: ✖ 2.74V

Correct Answer: ✔ -2.736 $\pm$ 5% (V)

Question 9

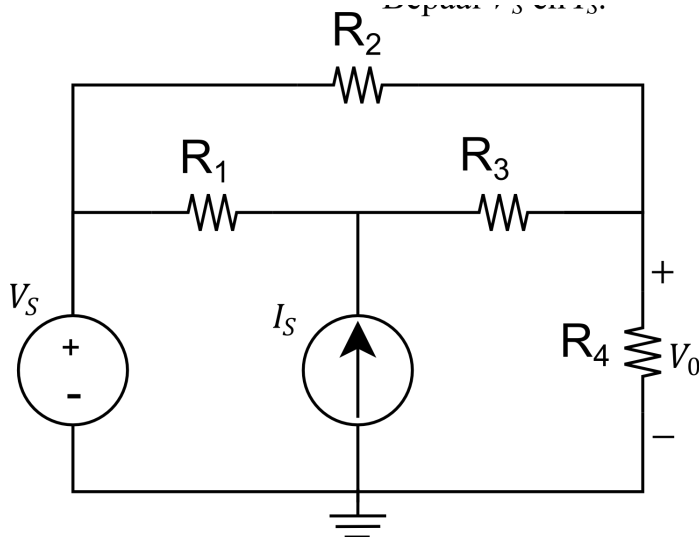
2 out of 2 points



The voltage source V_S contributes 8 V to V_0 , and the current source I_S contributes 4 V to V_0 .

Let $R_1 = 5 \, \Omega$, $R_2 = 15 \, \Omega$, $R_3 = 14 \, \Omega$ and $R_4 = 11 \, \Omega$.

Determine V_S



Selected Answer: ✔ 14.1V

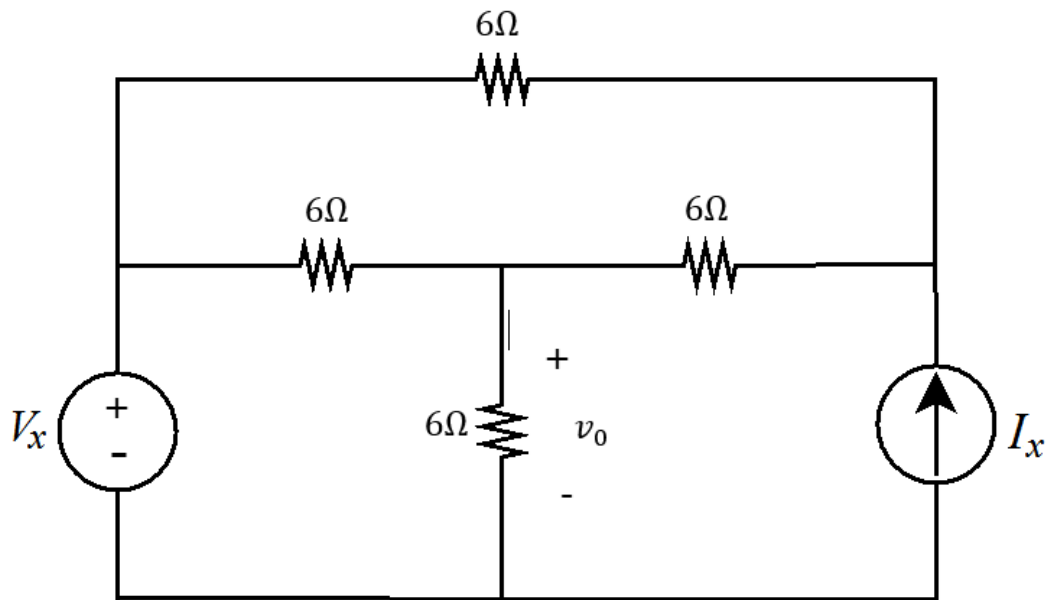
Correct Answer: ✔ 14.10 $\pm$ 5% (V)

Question 10

2 out of 2 points



Use superposition and calculate v_0 due to the $I_x = 1.27 \, \text{A}$ current source, given $V_x = 10 \, \text{V}$.



Selected Answer: ☒ 1.52V

Correct Answer: ☒ $1.52 \pm 5\%$ (V)

Monday, August 28, 2017 6:15:01 PM SAST

← OK